

EXHIBIT 23

Tower Semiconductor Patents – Key Insights and Stats

Tower Semiconductor has a total of 189 patents globally, out of which 148 have been granted. Of these 189 patents, more than 79% patents are active. The United States of America is where Tower Semiconductor has filed the maximum number of patents, followed by Japan and Europe. Parallely, The United States of America seems to be the main focused R&D center, and also USA is the origin country of Tower Semiconductor.

Tower Semiconductor was founded in the year 1993. Tower Semiconductor is an independent foundry providing semiconductor integrated circuits manufacturing & related design services and manufactures wafers using its technologies & production capability and the proprietary designs of its customers.

Do read about some of the most popular patents of Tower Semiconductor which have been covered by us in this article and also you can find Tower Semiconductor patents information, the worldwide patent filing activity and its patent filing trend over the years, and many other stats over Tower Semiconductor patent portfolio.

How many patents does the CEO of Tower Semiconductor have?

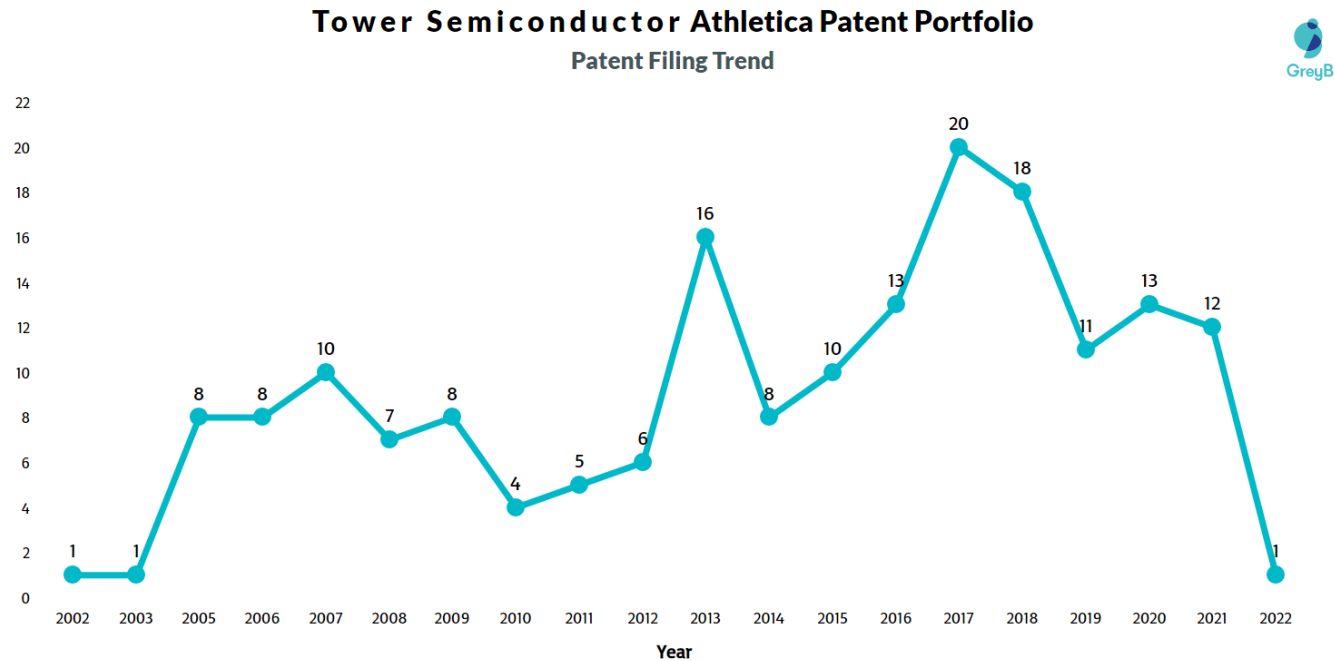
The CEO Russell C Ellwanger have 32 patents.

How many patents does Tower Semiconductor have?



Tower Semiconductor has a total of 189 patents globally. These patents belong to 135 unique patent families. Out of 189 patents, 151 patents are active.

How Many Patents did Tower Semiconductor File Every Year?



Note: Patent filing year and all the patents in the portfolio are considered to plot the filing trend, Data Range Based on Publication Year (2002-2023).

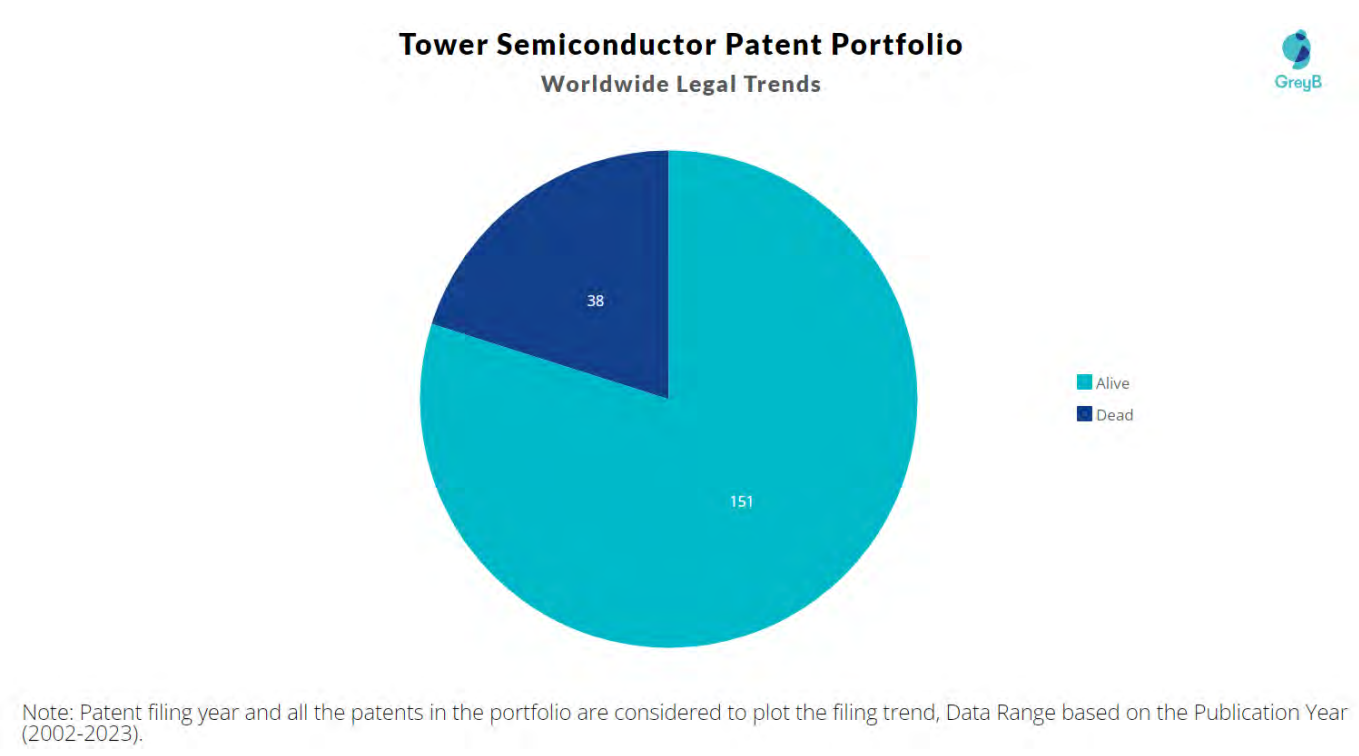
Are you wondering **why there is a drop in patent filing for the last two years?** It is because a patent application can take up to 18 months to get published. Certainly, it doesn't suggest a decrease in the patent filing.

Year of Patents Filing or Grant	Tower Semiconductor Applications Filed	Tower Semiconductor Patents Granted
2023	—	4
2022	1	11
2021	12	14
2020	13	22
2019	11	5
2018	18	11
2017	20	10

2016	13	13
2015	10	7
2014	8	4
2013	16	9
2012	6	3
2011	5	3

How many Tower Semiconductor patents are Alive/Dead?

Worldwide Patents



How Many Patents did Tower Semiconductor File in Different Countries?

Tower Semiconductor Patent Portfolio

Worldwide Patent Filing



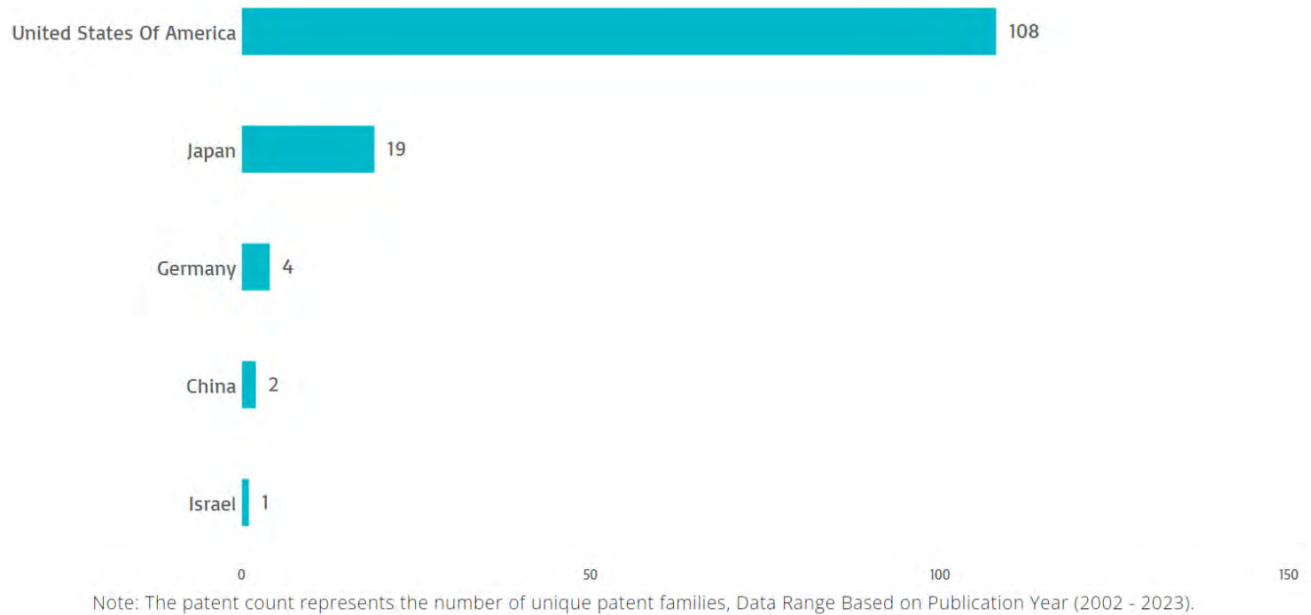
Note: The patent count is calculated by considering all patents in the portfolio, Data Range Based on Publication Year (2002 - 2023).

Countries in which Tower Semiconductor Filed Patents

Country	Patents
United States Of America	152
Japan	19
Europe	6
Germany	4
China	2

Where are the Research Centers of Tower Semiconductor Patents Located?

Tower Semiconductor Patent Portfolio R&D Centre



10 Best Tower Semiconductor Patents

[US6396741B1](#) is the most popular patent in the Tower Semiconductor portfolio. It has received 184 citations so far from companies like Macronix International Co, Saifun Semiconductor, and Spansion Israel Ltd.

Below is the list of 10 most cited patents of Tower Semiconductor:

Publication Number	Citation Count
US6396741B1	184
US8378407B2	146
US7609093B2	76
US9917104B1	67
US7358583B2	66
US6479963B1	61
US7227234B2	57
US7773138B2	53

US20130293986A1	50
US20030134476A1	50

How many inventions of other companies were rejected due to Tower Semiconductor patents?

The statistics below share strategic R&D insights. It tells the companies that were trying to protect inventions similar to Tower Semiconductor invention. They couldn't because Tower Semiconductor had protected those before them.

Examiners at the USPTO referred 78 Tower Semiconductor patents in 513 rejections (35 USC § 102 or 35 USC § 103 types).

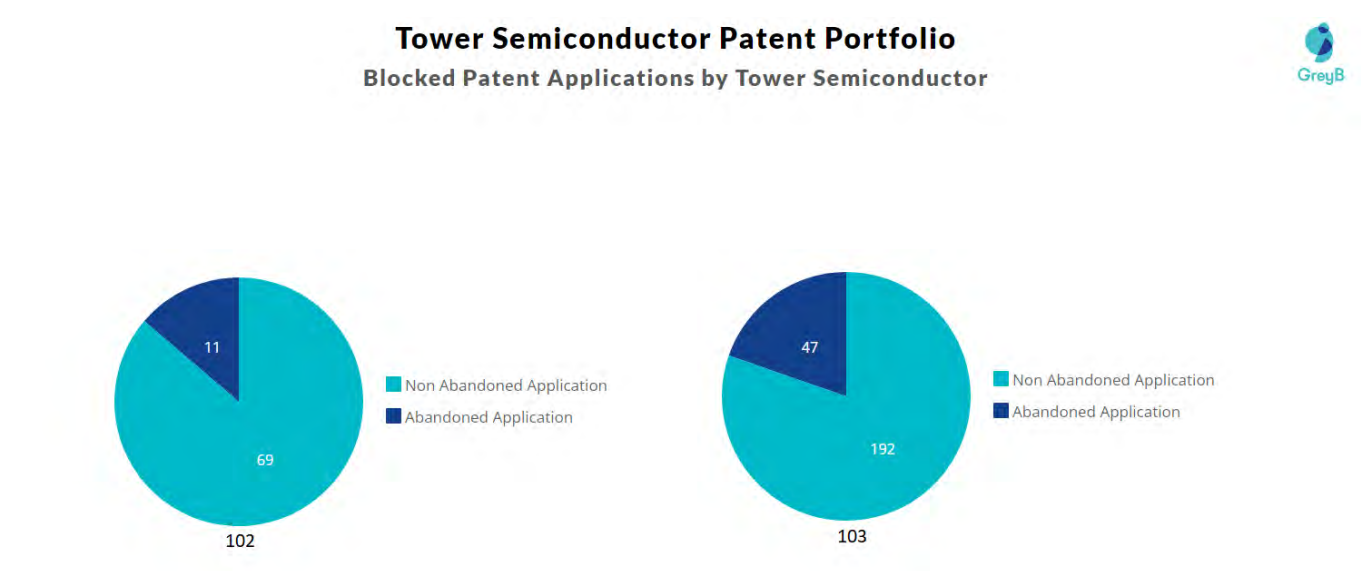
The top citing companies in the Tower Semiconductor patent portfolio are **TSMC, Texas Instruments, and Samsung**.

List of the Companies whose Patents were rejected citing Tower Semiconductor –

Company	Number of Patent Applications that faced Rejection Citing Tower Semiconductor Patents	Number of Rejections (102 & 103)
Tsmc	25	74
Texas Instruments	11	25
Samsung Electronics	10	17
Wilmington Trust	10	14
Sk Hynix	7	18
Sony Corporation	11	19
Renesas Electronics	7	15
IBM	6	6
Canon	6	13

Ememory Technology Incorporated	6	8
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Count of 102 and 103 Type Rejections based on Tower Semiconductor Patents



Top Tower Semiconductor Patents used in Rejecting Most Patent Applications

Patent Number	Count of Rejected Applications
US9330979B2	20
US9484454B2	13
US8378407B2	11
US8279328B2	11
US8004237B2	10

US7485941B2	9
US7358583B2	8
US8722484B2	8
US7575977B2	8
US7800156B2	7
US8520100B2	7
US9484800B2	7
US20070200055A1	6
US8253373B2	6
US7737390B2	6

What Percentage of Tower Semiconductor US Patent Applications were Granted?

Tower Semiconductor (Excluding its subsidiaries) has filed 202 patent applications at USPTO so far (Excluding Design and PCT applications). Out of these 180 have been granted leading to a **grant rate of 92.7%**.

Below are the key stats of Tower Semiconductor patent prosecution at the USPTO.

	Assignee	Group Art Units
Total Applications:	202	1407698
Granted Patents:	180	860191 (84.17%)
Grant Index	Impressive (92.78%)	
Abandoned/Rejected Applications:	14 (7.22%)	161739 (15.83%)
In-Process Applications:	7	385768
Average Grant Time:	2.21 Years	2.38 Years
Average Office Actions:	0.9	1.37

Which Law Firms Filed Most US Patents for Tower Semiconductor?

Law Firm	Total Applications	Success Rate
Bever Hoffman & Harms	136	92.54%

Reches Patents	15	100.00%
Shichrur & Co	11	100.00%
Birch Stewart Kolasch & Birch	9	100.00%
Maschoff Brennan	5	0%
Naschitz Brandes Amir & Co	4	100.00%
Kligler & Associates Patent Attorneys	3	100.00%
Prtsi Inc	3	66.67%
Eitan Law Group	2	100.00%
Noam Eshel	2	100.00%

What are Tower Semiconductor's key innovation segments?

Fill the form to submit your request for the data

Your name*

Your Work Email*

Company Name*

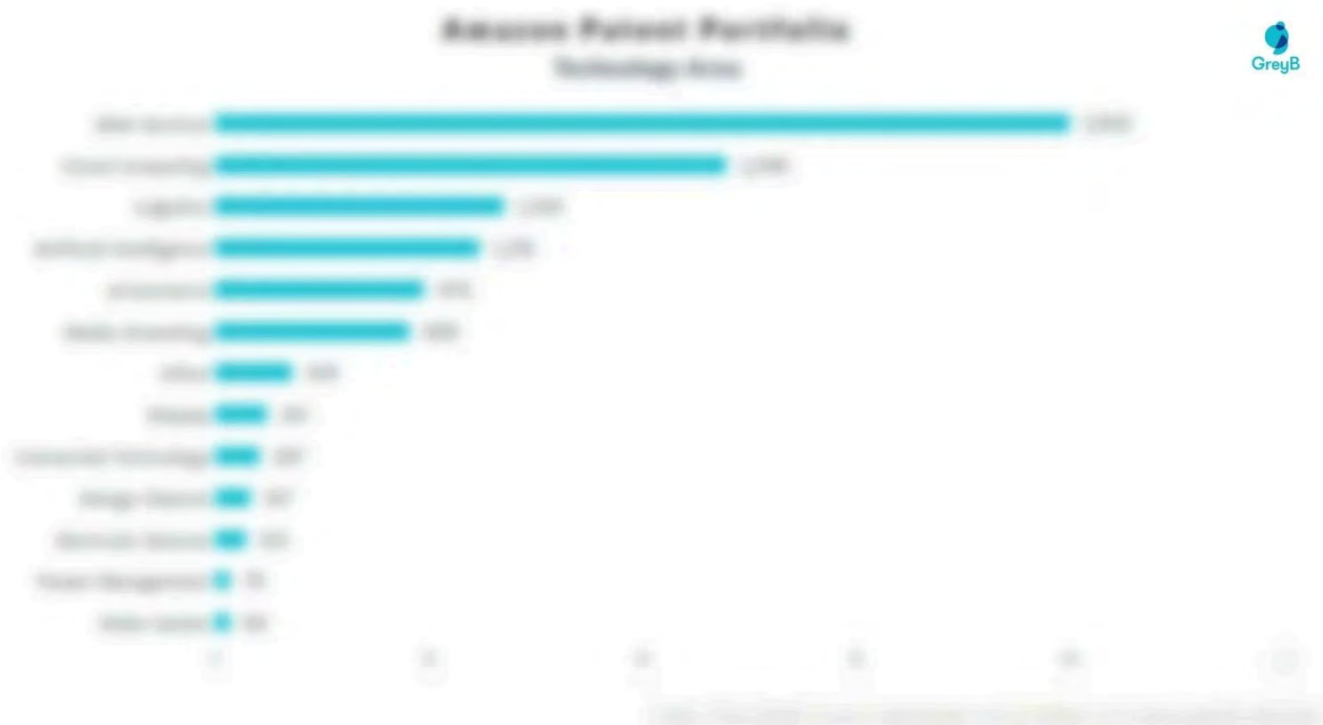
What best describes your profession? *

Please Select

▼

Request additional data

What Technologies are Covered by Tower Semiconductor?



The chart below distributes *patents filed by Tower Semiconductor in different countries on the basis of the technology protected* in patents. It also represents the markets where **Tower Semiconductor** thinks it's important to protect particular technology inventions.



R&D Focus: How has Tower Semiconductor's search focus changed over the years?



March 10, 2025

EXCLUSIVE INSIGHTS COMING SOON!

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INSIGHTS YOU MAY FIND INTERESTING

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Your name*

Email*

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Key technologies, leading companies,
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